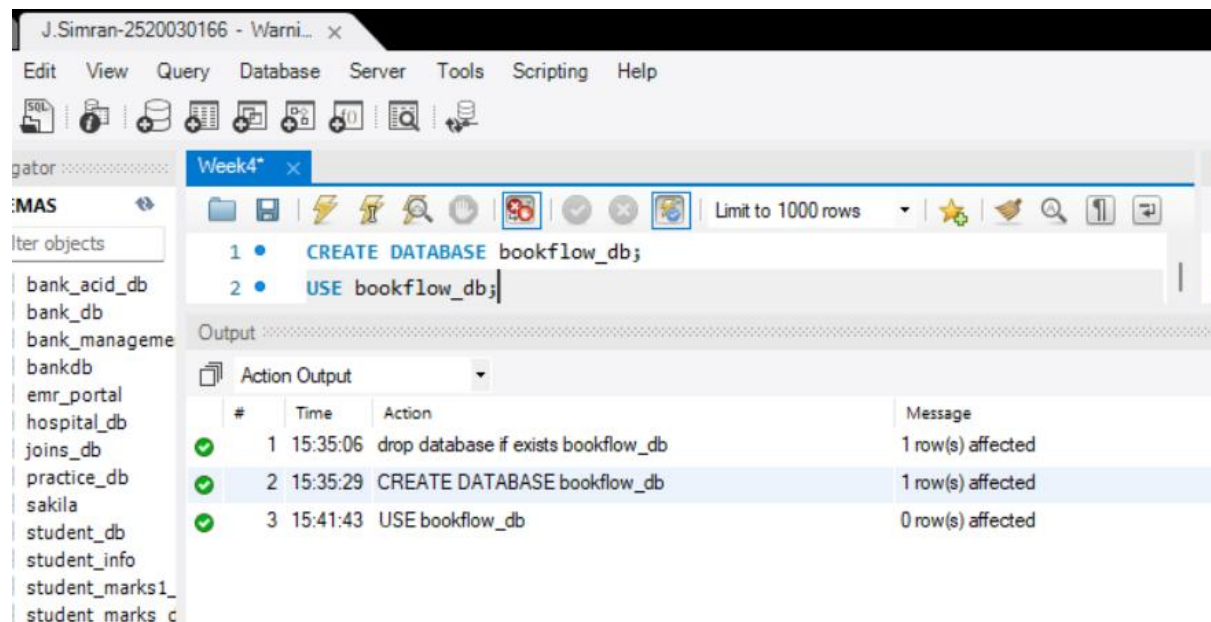


Week-4: BOOKFLOW INTELLIGENT CATALOG & API BLUEPRINT

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CREATE DATABASE:



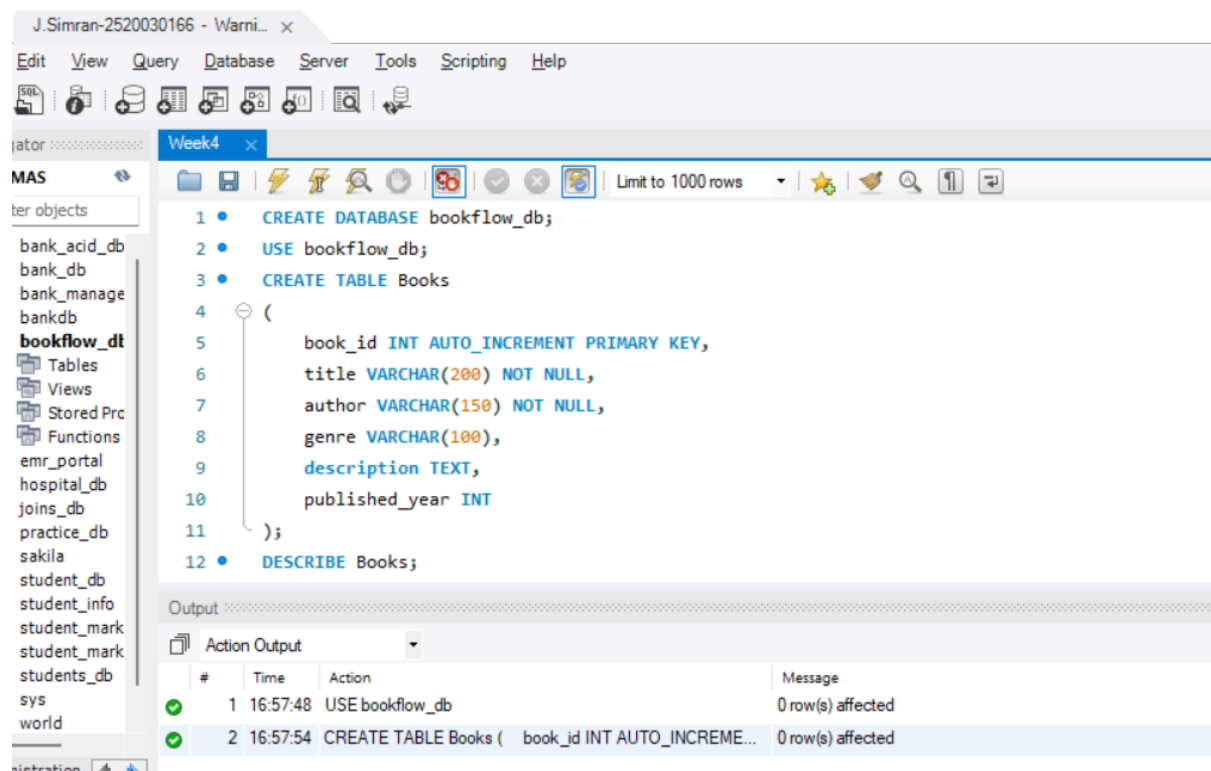
The screenshot shows the SQL Developer interface with the following SQL script in the editor:

```
1 • CREATE DATABASE bookflow_db;  
2 • USE bookflow_db;
```

The Output window displays the following Action Output:

#	Time	Action	Message
✓ 1	15:35:06	drop database if exists bookflow_db	1 row(s) affected
✓ 2	15:35:29	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 3	15:41:43	USE bookflow_db	0 row(s) affected

CREATE BOOKS TABLE:



The screenshot shows the SQL Developer interface with the following SQL script in the editor:

```
1 • CREATE DATABASE bookflow_db;  
2 • USE bookflow_db;  
3 • CREATE TABLE Books  
4 • (  
5 •     book_id INT AUTO_INCREMENT PRIMARY KEY,  
6 •     title VARCHAR(200) NOT NULL,  
7 •     author VARCHAR(150) NOT NULL,  
8 •     genre VARCHAR(100),  
9 •     description TEXT,  
10 •     published_year INT  
11 • );  
12 • DESCRIBE Books;
```

The Output window displays the following Action Output:

#	Time	Action	Message
✓ 1	16:57:48	USE bookflow_db	0 row(s) affected
✓ 2	16:57:54	CREATE TABLE Books (book_id INT AUTO_INCREME...	0 row(s) affected

CHECK TABLE:

The screenshot shows the SQL Developer interface with a SQL script in the main editor and its execution results in the Output pane.

SQL Script:

```
9      description TEXT,  
10      published_year INT  
11  );  
12  • DESCRIBE Books;  
13  • INSERT INTO Books  
14  (
```

Result Grid:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	

Output:

#	Time	Action	Message
✓ 1	16:57:48	USE bookflow_db	0 row(s) affected
✓ 2	16:57:54	CREATE TABLE Books (book_id INT AUTO_INCREME...	0 row(s) affected
✓ 3	17:01:01	SELECT * FROM Books LIMIT 0, 1000	0 row(s) returned
✓ 4	17:01:46	DESCRIBE Books	6 row(s) returned

INSERT BOOK DATA:

The screenshot shows the SQL Developer interface with a query window titled 'Week4*'. The query is as follows:

```
86
87 (
88     'Harry Potter and the Sorcerers Stone',
89     'J.K. Rowling',
90     'Fantasy',
91     'A young wizard begins his magical education.',
92     1997
93 ),
94
95 (
96     'Twenty Thousand Leagues Under the Seas',
97     'Jules Verne',
98     'Adventure',
99     'A fantastic underwater adventure aboard a mysterious submarine.',
100     1870
101 );
```

The left sidebar shows a tree view of database objects. The 'bookflow_dt' database is selected, and the 'Tables' folder is expanded. The 'Output' window at the bottom shows the execution results of the query.

#	Time	Action	Message
✓ 1	16:57:48	USE bookflow_db	0 row(s) affected
✓ 2	16:57:54	CREATE TABLE Books (book_id INT AUTO_INCREME...	0 row(s) affected
✓ 3	17:01:01	SELECT * FROM Books LIMIT 0, 1000	0 row(s) returned
✓ 4	17:01:46	DESCRIBE Books	6 row(s) returned
✓ 5	17:03:12	INSERT INTO Books (title, author, genre, descri...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

DISPLAY BOOKS:

J.Simran-2520030166 - Warni... x

Edit View Query Database Server Tools Scripting Help

Week4* x

MAS

ter objects

bank_acid_db
bank_db
bank_manage
bankdb
bookflow_dt
Tables
Views
Stored Proc
Functions
emr_portal
hospital_db
joins_db
practice_db
sakila
student_db
student_info
student_mark
student_mark
students_db
sys
world

```
104 title,  
105 author,  
106 genre,  
107 published_year  
108 FROM Books  
109 ORDER BY book_id;  
110 ALTER TABLE Books
```

Limit to 1000 rows

Result Grid

	book_id	title	author	genre	published_year
▶	1	The Martian	Andy Weir	Science Fiction	2011
	2	Dune	Frank Herbert	Science Fiction	1965
	3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
	4	Pride and Prejudice	Jane Austen	Romance	1813
	5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
	6	Foundation	Isaac Asimov	Science Fiction	1951
	7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
	8	Enders Game	Orson Scott Card	Science Fiction	1985

Books 3 x

Output

Action Output

#	Time	Action	Message
✓ 1	16:57:48	USE bookflow_db	0 row(s) affected
✓ 2	16:57:54	CREATE TABLE Books (book_id INT AUTO_INCREME...	0 row(s) affected
✓ 3	17:01:01	SELECT * FROM Books LIMIT 0, 1000	0 row(s) returned
✓ 4	17:01:46	DESCRIBE Books	6 row(s) returned
✓ 5	17:03:12	INSERT INTO Books (title, author, genre, descri...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 6	17:04:09	SELECT book_id, title, author, genre, publishe...	10 row(s) returned

ADD EMBEDDING COLUMN:

The screenshot displays a database management tool interface. The top menu bar includes 'Edit', 'View', 'Query', 'Database', 'Server', 'Tools', 'Scripting', and 'Help'. The left sidebar shows a tree view of database objects, with 'bookflow_dt' selected. The main query editor shows a SQL script with the following lines:

```
106     genre,  
107     published_year  
108 FROM Books  
109 ORDER BY book_id;  
110 • ALTER TABLE Books  
111 ADD COLUMN embedding JSON;  
112 • DESCRIBE Books;  
113 • UPDATE Books
```

Below the query editor, the 'Result Grid' tab is active, showing the structure of the 'Books' table:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	

The 'Output' tab is also visible, showing a log of database actions:

#	Time	Action	Message
✓ 1	16:57:48	USE bookflow_db	0 row(s) affected
✓ 2	16:57:54	CREATE TABLE Books (book_id INT AUTO_INCREME...	0 row(s) affected
✓ 3	17:01:01	SELECT * FROM Books LIMIT 0, 1000	0 row(s) returned
✓ 4	17:01:46	DESCRIBE Books	6 row(s) returned
✓ 5	17:03:12	INSERT INTO Books (title, author, genre, descri...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 6	17:04:09	SELECT book_id, title, author, genre, publishe...	10 row(s) returned
✓ 7	17:05:11	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 8	17:07:00	DESCRIBE Books	7 row(s) returned

INSERT MOCK VECTORS:

The screenshot shows a database management tool interface. The top menu bar includes Edit, View, Query, Database, Server, Tools, Scripting, and Help. Below the menu is a toolbar with various icons. The main window is titled "Week4*" and displays a list of SQL queries. The left sidebar shows a tree view of database objects, including bank_acid_db, bank_db, bank_manage, bankdb, bookflow_dt, Tables, Views, Stored Proc, Functions, emr_portal, hospital_db, joins_db, practice_db, sakila, student_db, student_info, student_mark, student_mark, students_db, sys, and world. The bottom section shows the "Output" tab with "Action Output" selected, displaying a table of execution results.

```
118 WHERE book_id = 2;
119 • UPDATE Books
120 SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
121 WHERE book_id = 3;
122 • UPDATE Books
123 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
124 WHERE book_id = 4;
125 • UPDATE Books
126 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
127 WHERE book_id = 5;
128 • UPDATE Books
129 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
130 WHERE book_id = 6;
131 • UPDATE Books
132 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
133 WHERE book_id = 7;
```

#	Time	Action	Message
✓ 11	17:08:35	UPDATE Books SET embedding = JSON_ARRAY(0.85, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 12	17:08:41	UPDATE Books SET embedding = JSON_ARRAY(0.80, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 13	17:08:45	UPDATE Books SET embedding = JSON_ARRAY(0.10, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 14	17:08:54	UPDATE Books SET embedding = JSON_ARRAY(0.15, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 15	17:09:00	UPDATE Books SET embedding = JSON_ARRAY(0.95, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 16	17:09:06	UPDATE Books SET embedding = JSON_ARRAY(0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 17	17:09:26	UPDATE Books SET embedding = JSON_ARRAY(0.88, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 18	17:09:33	UPDATE Books SET embedding = JSON_ARRAY(0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 19	17:09:39	UPDATE Books SET embedding = JSON_ARRAY(0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...

VERIFY ALL VECTORS:

J.Simran-2520030166 - Warni... x

Edit View Query Database Server Tools Scripting Help

ator Week4 x

er objects

bank_acid_db
bank_db
bank_manage
bankdb
bookflow_dt
Tables
Views
Stored Proc
Functions
emr_portal
hospital_db
joins_db
practice_db
sakila
student_db
student_info
student_mark
student_mark
students_db
sys
world

143 • SELECT
144 book_id,
145 title,
146 embedding
147 FROM Books
148 ORDER BY book_id;
149 • SELECT

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

	book_id	title	embedding
▶	1	The Martian	[0.90, 0.80, 0.20]
	2	Dune	[0.85, 0.75, 0.25]
	3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
	4	Pride and Prejudice	[0.10, 0.20, 0.90]
	5	The Great Gatsby	[0.15, 0.30, 0.80]
	6	Foundation	[0.95, 0.65, 0.20]
	7	The Hobbit	[0.30, 0.75, 0.90]
	8	Enders Game	[0.88, 0.85, 0.15]
	9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
	10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]
•	NULL	NULL	NULL

Books 5 x A3

Output

Action Output

#	Time	Action	Message	Dur
✓ 15	17:09:00	UPDATE Books SET embedding = JSON_ARRAY(0.95, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...	0.03
✓ 16	17:09:06	UPDATE Books SET embedding = JSON_ARRAY(0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...	0.03
✓ 17	17:09:26	UPDATE Books SET embedding = JSON_ARRAY(0.88, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...	0.03
✓ 18	17:09:33	UPDATE Books SET embedding = JSON_ARRAY(0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...	0.03
✓ 19	17:09:39	UPDATE Books SET embedding = JSON_ARRAY(0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...	0.03
✓ 20	17:14:30	SELECT book_id, title, embedding FROM Books ...	10 row(s) returned	0.00

Info Session

SEARCH VECTOR:

L2 DISTANCE

The screenshot shows a database management tool interface with a SQL editor and a results pane. The SQL editor contains a query to calculate the L2 distance for books, and the results pane displays the output of this query.

SQL Query:

```
176 ) - 0.20,  
177 2  
178 )  
179 ),  
180 4  
181 ) AS l2_distance  
182 FROM Books  
183 WHERE embedding IS NOT NULL  
184 ORDER BY l2_distance ASC  
185 LIMIT 3;  
186 SELECT  
187  
188 book_id
```

Result Grid:

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	Dune	Frank Herbert	Science Fiction	0.0866

Action Output:

#	Time	Action	Message
✓ 16	17:09:06	UPDATE Books SET embedding = JSON_ARRAY(0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 17	17:09:26	UPDATE Books SET embedding = JSON_ARRAY(0.88, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 18	17:09:33	UPDATE Books SET embedding = JSON_ARRAY(0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 19	17:09:39	UPDATE Books SET embedding = JSON_ARRAY(0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
✓ 20	17:14:30	SELECT book_id, title, embedding FROM Books ...	10 row(s) returned 1 row(s) af Rows matc
✓ 21	17:17:00	SELECT book_id, title, author, genre, ROUN...	3 row(s) returned

COSINE SIMILARITY:

J.Simran-2520030166 - Warni... x

Edit View Query Database Server Tools Scripting Help

gator Week4*

filter objects

- bank_acid_db
- bank_db
- bank_manage
- bankdb
- bookflow_dt
 - Tables
 - Views
 - Stored Proc
 - Functions
- emr_portal
- hospital_db
- joins_db
- practice_db
- sakila
- student_db
- student_info
- student_mark
- student_mark
- students_db
- sys
- world

Initialization

Information

```
238 +
239 POW(0.80, 2)
240 +
241 POW(0.20, 2)
242 )
243 ),
244 4
245 ) AS cosine_similarity
246 FROM Books
247 WHERE embedding IS NOT NULL
248 ORDER BY cosine_similarity DESC
249 LIMIT 3;
```

Result Grid

	book_id	title	author	genre	cosine_similarity
▶	1	The Martian	Andy Weir	Science Fiction	1
	2	Dune	Frank Herbert	Science Fiction	0.9986
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 7 x

Output

Action Output

#	Time	Action	Message
✓ 17	17:09:26	UPDATE Books SET embedding = JSON_ARRAY(0.88, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning.
✓ 18	17:09:33	UPDATE Books SET embedding = JSON_ARRAY(0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning.
✓ 19	17:09:39	UPDATE Books SET embedding = JSON_ARRAY(0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Warning.
✓ 20	17:14:30	SELECT book_id, title, embedding FROM Books ...	10 row(s) returned
✓ 21	17:17:00	SELECT book_id, title, author, genre, ROUN...	3 row(s) returned
✓ 22	17:19:16	SELECT book_id, title, author, genre, ROUN...	3 row(s) returned

ct Info Session